

CHEMICAL ENGINEERING 317
INTRODUCTION TO CHEMICAL ENGINEERING ANALYSIS
Spring 2026, Unique # 16455

Lectures: Monday, Wednesday, Friday from 8:00 am until 8:50 in CPE 2.214
Recitation: Fridays 1:00 – 3:00 pm in ASE 1.126

PROFESSOR:

Benny D. Freeman

Office: EER 4.614B Office hours: Monday, Wednesday, Friday 9 – 10 am, EER 4.702
and by appointment

Phone: (512)232-2803 Email: benny.freeman@utexas.edu

Mobile: (512)626-6612

DR. FREEMAN'S ADMINISTRATIVE ASSISTANT:

Carol Assadourian

Phone: (512)471-4235 Email: carol.assadourian@austin.utexas.edu

TEACHING ASSISTANT:

Busra Abdulkadiroglu (CPE 3.416)

Email: babdulkadiroglu@utexas.edu

Phone: (919)951-8095

Office Hours/Location: 8:30 – 10:00 Tues, Thurs, EER 4.702

GRADER:

Makena Kao Office Hours/Location: 12:00 – 2:30 pm Tuesdays, EER 4.704

Contact: makena.kao104@gmail.com

Phone: (817)312-8684

TUTOR:

Joseph Schiller Office Hours: 5:00 – 7:00 pm Mondays and Tuesdays, EER 4.702

Contact: jds8862@my.utexas.edu

Phone: (281)475-6435

Course Goals and Objectives: To introduce students to the principles and calculation techniques used in the field of chemical engineering.

- To acquaint students with the fundamentals of material and energy balances as applied to chemical engineering.
- To acquaint students with efficient methods of **problem solving** so that they can effectively solve problems they will encounter after leaving school.
- To offer practice in defining problems, collecting data, analyzing the data and breaking it down.
- To help a student decide if he or she has chosen the right field.

Knowledge, Abilities, and Skills Students Should Have Before Entering the Course:

- One year of freshman mathematics (algebra, solution of simultaneous equations, quadratic equations, basic differentiation and integration, trigonometry and basic geometry [area & volume of simple shapes]). *M408C, M408D*
- Chemistry (stoichiometry, concept of mole, molecular weight, concentration, molarity, molality, etc., enthalpy of reaction, ideal gas law, pressure, absolute temperature). *Chem 302*
- Basic knowledge of statics and dynamics in physics (concepts of force, work, pressure, kinetic energy, potential energy). *Phys 303 or high school physics*
- Time management skills.

- Spreadsheet calculations for solving systems of linear equations, function minimization, and plotting.
ChE 210

Knowledge, Abilities, and Skills Students Should Gain From This Course:

- Familiarity with chemical process equipment.
- Ability to work in groups.
- Exposure to systems analysis and the concept of unit operations.
- Exposure to library and reference data sources and the need to use them as an engineer.
- Ability to make calculations using different sets of units, including the appropriate use of the gravitational conversion factor and gauge vs. absolute pressure.
- Ability to write and balance chemical reaction equations. Ability to identify the excess and limiting reactants and calculate the percent conversion and yield of a reaction.
- Ability to define steady state and distinguish between open and closed systems.
- Ability to solve material balances for systems with or without a chemical reaction.
- Ability to solve mass balances involving interconnected units with recycle, bypass, and purge.
- Ability to define an ideal gas and make calculations using the perfect gas equation.
- Ability to apply Dalton's and Amagat's laws for ideal gas mixtures.
- Ability to apply equations of state (e.g., van der Waals' equation) for real gas calculations and define conditions for using equations of state.
- Ability to make calculations for non-ideal gaseous mixtures, including prediction of p-V-T behavior, using equations of state and compressibility factors.
- Ability to define vapor pressure, saturation, equilibrium, triple point, dew point, partial saturation (humidity), etc. Know the difference between wet and dry bulb temperatures.
- Ability to apply Raoult's and Henry's laws for multicomponent systems.
- Ability to apply the phase rule.
- Ability to define heat, work, energy, enthalpy, etc.
- Ability to make estimations of heat capacity and to calculate enthalpy changes, with and without phase changes.
- Ability to use the steam tables.
- Ability to solve energy balances for closed and open systems, with and without chemical reaction.
- Ability to solve simple combined material and energy balances (with and without chemical reaction).
- Ability to distinguish and make calculations for adiabatic and non-adiabatic processes.
- Ability to apply spreadsheet calculations to mass and energy balance problems.

Chemical Engineering Program Outcomes Achieved:

Program Outcomes are a list of knowledge, skills, and abilities every Chemical Engineering student should have at the time of graduation. The Chemical Engineering Department at UT-Austin has defined three such outcomes as listed on the departmental web site:

<https://che.utexas.edu/prospective-students/undergraduate-students/about-the-program/accreditation/>

1. Become leading professionals who advance chemical engineering practice and knowledge in multiple fields, such as energy, materials, environmental and systems engineering, electronics, biotechnology, human health, and education.
2. Continue to educate themselves as their needs, interests, and circumstances dictate.
3. Become ethical and productive engineers, who recognize and acknowledge the local and global impacts of engineering technology on humans and the environment.

Chemical Engineering Student Outcomes Achieved (ABET Student Outcome 1):

Graduation from the Chemical Engineering degree program at The University of Texas at Austin signifies attainment of Student Outcomes. ChE 317 contributes to the following Student Outcome:

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics. Complex engineering problems are those having one or more of the following characteristics: “involving wide-ranging or conflicting technical issues, having no obvious

solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.”

Impact on Subsequent Courses in Curriculum:

Virtually every subsequent ChE course requires material and energy balance concepts. In addition, properties of materials (gas laws, vapor pressures, etc.) are important in ChE 322, 354, 350, 363, & 473K.

Required Text: *Basic Principles and Calculations in Chemical Engineering*, 9th Edition by Himmelblau & Riggs (ISBN: 9780137327171). There are several international editions of this text available. However, these international editions do not have the same content as the US edition. This class will only use the US edition of the textbook. We cannot offer support for versions other than the US version. Additionally, the 9th edition was just published in July 2022, and it contains significant differences from the 8th Edition, which we used in prior years. Homework problems and course materials are different in the 9th Edition. We cannot offer support for any editions other than the 9th Edition.

Class Website: Canvas (<https://canvas.utexas.edu/>)

Outline

INTRODUCTION TO ENGINEERING ANALYSIS

- Problem Definition and “How to get Started” on a General Problem
- Dimensions, Units & Conversions

MATERIAL BALANCES

- Basic Types: mass, elemental & chemical component
- Single Unit: reactive & nonreactive
- Multi-unit non-reactive processes: bypass, recycle & purge
- Multi-unit processes with reactions: bypass, recycle & purge

PROPERTIES OF MATERIALS

- Ideal gases: pure & mixed with applications to material balances
- Real gases: pure & mixed with applications to material balances (focus on z factors but introduce equations of state and trial & error solutions)
- Vapor pressure: Antoine, Clausius-Clapeyron, Cox charts & steam tables
- Gas–Vapor Mixtures: relative saturation, relative humidity, dew point temperature, dew point pressure, condensation, saturation and drying problems
- Vapor–Liquid Equilibrium: Raoult’s Law

ENERGY BALANCES

- Calculations of potential, internal, kinetic and enthalpy changes. Use of polynomial C_p , consideration of phase changes. Use of thermodynamic charts — e.g., Steam Tables
- Non-reactive Energy Balances: open steady state turbines, heat exchangers, throttle valves & filling & emptying of vessels
- Reactive Energy Balances: mostly open steady state systems, including multiple reactions with arbitrary feed and conversions and temperatures

LECTURES:

Please be punctual. Attendance is mandatory (see **Quizzes** below). Most lectures will be based on reading assignments and example problems that you *must* complete prior to coming to class. You will be asked questions based on that assignment.

RECITATION:

Please be punctual. Attendance is mandatory (see **Quizzes** below). The exams for the course will be held in recitation. Many of these sessions will be used to discuss homework problems and provide additional examples of problems that we are discussing in class. You must attend the recitation section for which you are registered. The Teaching Assistant will be running the recitation sections.

GRADING:

	Weight (%)
Homework #¹	10
In-Class Quizzes #²	5
Tests #³	60
Final Exam	25

- #1 **Homework:** Homework will generally be assigned each week. **The homework will be due at the beginning of class the week after it is assigned.** Late homework will **not** be accepted. Homework must be neat and presented in the standard ChE format. Instructions for preparing homework in this format are available on the Canvas website for this course. **Homework may be handed in by individuals, but you are encouraged to submit homework assignments in groups of two or three.** When working in groups on the homework, you should set up each problem by yourself and understand the approach to solving the problem before meeting with the group. This advance preparation will familiarize you with each problem. Moreover, one of the most critical steps in problem solving is arriving at an initial understanding of the solution strategy. These skills must be developed individually to a sufficient level to permit you to solve problems on exams and quizzes, where your individual skills in the course material will be assessed. You have one week after the homework is returned to you to request re-grading. We have the prerogative to regrade the entire homework, so that your final grade may be corrected up or down depending on the regrade of the entire problem set. No requests for regrades will be accepted later than one week after the graded homework assignment is returned to you.

Homework will be submitted electronically via Canvas. You may prepare your assignment electronically (e.g., on a tablet) and submit it electronically. You may also prepare your assignment on paper and scan it to a pdf file to submit via Canvas. If you work in a group, please put the name of all group members on the assignment, and only one group member needs to upload the assignment to Canvas. I typically use the free version of [Adobe Scan](#) or [Genius Scan](#) to scan and upload documents using my phone or tablet, but there are a number of good scanning applications that you may choose from for this purpose.

Reading Assignments: You will be given a reading assignment as well as a homework assignment on a weekly basis. You will be expected to have read the assigned sections of the text *prior to coming to the next lecture*. You should use the *Key Ideas*, *Key Words*, *Self-Assessment Test*, *Thought Problems*, and *Discussion Questions* at the end of each section in the text to make sure you understood the reading assignment *prior to coming to the next lecture*. Additionally, example problems will be assigned, and we will discuss these problems at the following class meeting. You are expected to have prepared a solution to these problems before we go over them in class.

- #2 **Quizzes:** There will be several unannounced quizzes of one or two questions each held during lectures and recitation sessions. The quizzes will be based on material covered in reading assignments, classroom discussions, and homework. You have one week after the quiz is returned to request re-grading. Please note that if you are not in class or recitation when the quiz is returned, it is your responsibility to contact the professor and/or teaching assistant to get your quiz. The clock on requesting regrading starts when the graded quiz is formally returned to the class. All quizzes count (see **Grading Scheme** above.) The quizzes will not be equally weighted; they will be weighted according to their difficulty.

#3 **Examinations:** Three tests are scheduled in recitation on **Friday, February 14, March 29, and April 24.** You have one week after the test is returned to you to request re-grading. Please note that if you are not in class or recitation when the test is returned, it is your responsibility to contact the professor and/or teaching assistant to get your test. The clock on requesting regrading starts when a graded test is formally returned to the class. All tests count towards your final grade. The tests will not be equally weighted in determining the final grade; they will be weighted according to their difficulty. **The final examination will be Friday, May 1, 8:00 am – 10:00 am** (<https://registrar.utexas.edu/schedules/262/finals>). The exams will not be equally weighted; they will be weighted according to their difficulty.

The policy on quiz and exam regrades is: we will go over exam and quiz solutions in class or in recitation. If you feel that your test has not been graded correctly after reviewing the solutions, you must hand back the test with a written explanation of the oversight within one week after the exam or quiz is returned to you. We have the prerogative to regrade the entire test, so that your final grade may be corrected up or down depending on the regrade of the entire exam or quiz.

ABSENCE POLICY: No excuses for missed tests, quizzes or final exam will be accepted other than written medical excuses *or* letters written on university letterhead for UT-related school activities. If you miss a test, quiz or final exam, a grade of zero will be recorded for that test, quiz or final exam.

Other Information:

If you are a student with a disability, or think you may have a disability, and need accommodations please contact Services for Students with Disabilities (SSD). You may refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with SSD, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations.

Academic dishonesty: The University of Texas has an honor code, and you are expected to abide by it. You may find a copy of it at: <http://deanofstudents.utexas.edu/conduct/>. Please do not hesitate to ask questions of the instructor if you have any doubts as to whether any action or behavior on your part or the part of others constitutes academic dishonest.

Accommodations for religious holidays: By UT Austin policy, you must notify me (the instructor) of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class, an examination, a work assignment, or a project to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Recommendations regarding emergency evacuation from the Office of Campus Safety and Security, 512-471-5767, <http://www.utexas.edu/safety/>

- Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.
- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors.
- Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- Behavior Concerns Advice Line (BCAL): 512-232-5050

- Link to information regarding emergency evacuation routes and emergency procedures can be found at:
<http://emergency.utexas.edu/preparedness>

COVID-19 SAFETY: UT maintains an active effort to provide updated information and access to resources (e.g., testing, vaccination, etc.). You may find that information here: <https://protect.utexas.edu/>